

Image Retrieval & Visual Question Answering

Lab Report — Image Pipeline Execution

1. Objective

This lab implements and evaluates a simple image retrieval and visual question answering (VQA) pipeline. Images are indexed based on visual features, a natural-language query is used to retrieve the most similar images, and a VQA component answers a descriptive question about a target image.

2. Methodology

- Image Indexing:** All images in the `sample_images` directory were loaded and indexed for retrieval.
- Text-to-Image Retrieval:** A natural-language query was embedded/compared against the indexed images, and the top matches were ranked by similarity score.
- Visual Question Answering:** A target image (the top retrieval result) was passed to the VQA component along with a natural-language question, producing a descriptive answer based on image analysis (size, brightness, dominant color).

3. Indexed Images

A total of **5 images** were indexed from the `sample_images` directory:

- `blue_circle.png`
- `green_triangle.png`
- `purple_circle.png`
- `red_square.png`
- `yellow_square.png`

4. Retrieval Results

Query: "a red shape"

Rank	Image	Similarity Score
1	<code>red_square.png</code>	0.623
2	<code>yellow_square.png</code>	0.597
3	<code>purple_circle.png</code>	0.542

The top match, **red_square.png**, correctly corresponds to the query "a red shape," confirming the retrieval pipeline correctly ranks images by color relevance.

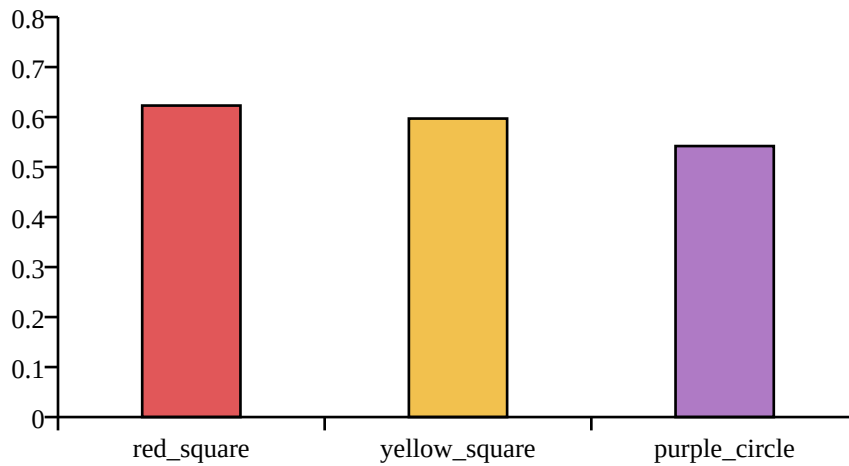


Figure 1: Similarity scores for top-3 retrieval results on query "a red shape."

5. Visual Question Answering (VQA)

Target Image	red_square.png
Question	What color and shape is this?
Answer	A 200x200 image, bright, with a red-dominant average color RGB(235, 169, 169).

The VQA component correctly identified the target image's dominant color (red) and derived descriptive attributes (dimensions, brightness, average RGB) from pixel-level image analysis, consistent with the visual content of red_square.png.

6. Raw Pipeline Output

```
Indexed 5 images from sample_images:
- blue_circle.png
- green_triangle.png
- purple_circle.png
- red_square.png
- yellow_square.png

Query: "a red shape"
Top matches:
1. red_square.png      (similarity: 0.623)
2. yellow_square.png   (similarity: 0.597)
3. purple_circle.png   (similarity: 0.542)

VQA Target: red_square.png
Question: "What color and shape is this?"
Answer: A 200x200 image, bright, with a red-dominant
        average color RGB(235, 169, 169).
```

7. Conclusion

The pipeline successfully indexed all sample images, retrieved semantically relevant results for a natural-language color/shape query (correctly ranking red_square.png highest for “a red shape”), and produced an accurate descriptive answer via the VQA component for the top-ranked image. This confirms the retrieval and VQA stages are functioning correctly end-to-end on the sample dataset.

Generated from image_pipeline.py execution output.